

Curriculum Overview

The central premise of this curriculum is that **humans are an intrinsic part of the natural world**. All social, political, and institutional systems exist in an active, continuous relationship with ecological systems.

Overview of the Modules

#	Lesson Name	Length	Description
Module 1: Exploring Why We Care			
1.1	How is Our Ocean Changing? Lesson Plan Lesson Slides	Two sessions of 50 minutes	Students begin exploring how climate change is affecting Southern California’s ocean and how different tools — including generative AI (GenAI) models — can help us investigate scientific questions. Students start by reflecting on their personal experiences with the ocean. They then investigate how climate change is impacting Southern California’s ocean by comparing two digital tools: a search engine and a GenAI platform, such as ChatGPT. They are introduced to the CLEAR framework for prompting GenAI and then work in small groups to use both tools to research ocean change and compare the differences in the information they find.
1.2	Planning Community Interviews Lesson Plan Lesson Slides	50 minutes	Student teams get ready to explore community perspectives on the ocean by planning interviews with people that they know. These interviews will later serve as the foundation for students to build a model that shows how climate change affects the ocean and human communities. Students begin by reflecting on how social scientists use interviews to understand community perspectives. After identifying the purpose of their interviews, they work in small teams to select questions from a question bank and plan their approach. To prepare, they watch a teacher-facilitated demonstration of ineffective interview

			techniques and discuss strategies for conducting successful interviews. By the end of the lesson, students will be ready to engage with community members and gather meaningful insights.
1.3	Analyzing Interview Data Lesson Plan Lesson Slides	50 minutes	Students analyze the notes they take during the interviews to identify key themes and patterns in how people in their community connect with the ocean. They begin by reviewing their notes to categorize responses into three main areas: past experiences with the ocean, reasons they care about the ocean, and perspectives on how the ocean is changing. Using thematic posters, students sort and organize ideas to visualize commonalities across interviews. As a class, they then discuss emerging patterns, exploring the shared values, concerns, and lived experiences that shape their community's relationship with the ocean. Finally, students consider how these insights might inform their own understanding of ocean change and its impact on their community.
Module 2: Building a Model			
2.1	Building Our Model Lesson Plan Lesson Slides Review SageModeler Instruction	Two sessions of 50 minutes	Students work together in small teams to develop a model showing their initial ideas about how climate change will affect Southern California's ocean. Drawing from their analysis of community interviews, student teams brainstorm the components, processes, and human interactions that play a role in the ocean. They review their interview analysis and select one idea that their community values about the ocean to focus on. They then use SageModeler to begin developing a model that shows how they predict that climate change might affect the value. Finally, they reflect on how they can use GenAI chatbots to identify additional perspectives and ideas to incorporate

			into their model.
2.2	Refining Our Model with AI Lesson Plan Lesson Slides Review AI Chatbot Platform	50 minutes + Extended/ Optional Take Home Activity (15 minutes)	Student teams refine their initial models of how climate change impacts the values Southern California communities place on the ocean. Students begin by analyzing their initial model to identify what questions they have and gaps in information. They then use GenAI chatbots to gather additional information on other perspectives, ideas, and data that helps support the gaps in their models. Finally, student teams use the gathered information and integrate the new perspectives into their initial model.
Module 3: Refining Our Model			
3.1	The Carbon Cycle Lesson Plan Lesson Slides	50 minutes	Students begin exploring how carbon moves through Earth's systems, the hydrosphere, atmosphere, geosphere, and biosphere, and explore how human activities can influence this cycle. Students begin their reflections on the sources of carbon dioxide within their communities with the goal of understanding how these sources influence global processes.
3.2	Carbon Dioxide & Weather Lesson Plan Lesson Slides Review PhET Instruction (Day 2) Greenhouse Effect: Waves Simulation	Two sessions of 50 minutes	Students investigate how rising atmospheric carbon levels influence weather patterns by trapping heat and driving changes to the Earth's climate. Building on their model from Lesson 3.1, students focus on understanding how human activities influence the greenhouse effect.

3.3	CO₂ and Sea Level Rise/ Coastal Erosion Lesson 3.3/3.4 Introduction Slides Lesson Plan Lesson Slides Review Our Coast Our Future Instruction	Two sessions of 50 minutes	During this lesson, students investigate how rising atmospheric carbon dioxide can lead to sea level rise and coastal erosion through changes in other processes, such as global warming and the frequency and intensity of storms. They also explore how these changes can affect the value of their community in Southern California's ocean.
3.4	Ocean Acidification Lesson 3.3/3.4 Introduction Slides Lesson Plan Lesson Slides Review SageModeler Instruction and Food Web Model	Two sessions of 50 minutes	Students investigate how rising atmospheric carbon dioxide levels can catalyze changes in ocean chemistry, particularly by increasing the rate of a process known as ocean acidification. They then explore how this rate of change can impact marine organisms made of calcium carbonate shells and eventually the ocean food web. They also explore how these changes could affect human reliance on ocean resources, such as fisheries and coastal livelihoods.
3.5	Update Our Model	50 minutes	Students revisit their models from Module 2 that share how their communities value the ocean. Students use the evidence they gathered from their investigations to expand and refine the processes in their model, such as sea level rise, coastal erosion, ocean acidification, or global

	Lesson Plan Lesson Slides SageModeler Examples		warming, and how human impacts drive these processes to change over time. After refining the models, students begin to prepare for Module 4 by planning what they want to communicate to their community as a potential solution or action they can take to mitigate the effects of climate change on the values they place on Southern California's ocean.
Module 4: Designing a Solution			
4.1	Identifying Solutions Lesson Plan Lesson Slides Review Teacher Rubric and Student Checklist	50 minutes	Students get ready to communicate to the public about local climate change issues through a science communication product. Students will work in groups to identify the message they want to communicate, drawing from the models they have created and revised in previous modules. Students will be introduced to key principles of science communication and use GenAI tools to practice these principles.
4.2	Choose Our Audience and Plan Our Approach Lesson Plan Lesson Slides Review Teacher Rubric and Student Checklist	50 minutes	Students continue working on their science communication products. They will build the profiles of their target audience and choose their communication approach. By the end of the lesson, they should have a preliminary science communication plan and a defined target audience to test in Module 4.3.

4.3	Build a Chatbot to Test Our Message Lesson Plan Lesson Slides Review Teacher Rubric and Student Checklist	50 minutes	Students will test the science communication plan they have created in Module 4.2. They will do so by building a chatbot using GenAI to represent their target audience. They will test the communication plan with the chatbot. Students will gather feedback from the chatbots to help them refine their products in Module 4.4.
4.4	Put Our Plan into Action Lesson Plan Lesson Slides Review Teacher Rubric and Student Checklist	Two sessions of 50 minutes	Students will refine their science communication products based on the feedback they receive from Module 4.3. They will build the products (e.g., with Canva) to convey their climate issue, their key message, and supporting evidence to the chosen audience. By the end of the lesson, students should have a more polished draft of their products that can be presented to the target audience.

Unit Learning Objectives

Science

By the end of this unit, students will be able to...	You can assess this using...
1. Analyze community interview data and other information (from web search, chatbot interactions) to identify themes about why local people care about the ocean.	Interview themes in lesson 1.3. Chatbot interactions and classroom discussion in lessons 2.1, 2.2.
2. Develop a computational representation showing how human-caused changes to Southern California's ocean over space and time will impact something that their community cares about.	Model created with SageModeler in lesson 2.1.
3. Create a quantitative model showing how imbalances in carbon cycling through Earth's systems are causing Southern California's ocean to change over space and time.	Model created with data analysis in Module 3.
4. Use computational representations to support a claim about how Southern California's ocean will change in the future.	Model created with data analysis in Module 3.
5. Create a science communication product conveying how a specific audience can take action to address climate change.	Science communication artifact in Module 4.

Computer Science (AI)

By the end of this unit, students will be able to...	You can assess this using...
6. Understand how GenAI works by adjusting their natural representations and predicting the output.	Class activity in lessons 1.1.
7. Reflect on the advantages, disadvantages, and ethics of using GenAI.	Class discussion in lessons 1.1, 2.2.
8. Understand how to prompt text-based GenAI tools to acquire information, give feedback, and communicate about their models.	Class discussion and prompt practice in lessons 1.1, 2.2. Science communication artifact in Module 4.
9. Validate GenAI's responses for accuracy and relevance.	Class discussion and reflection in lessons 1.1, 2.2, Module 4.
10. Create a GenAI chatbot to represent an audience's perspective and use it to test and refine their science communication messages.	GenAI prompts in Module 4

Standards Alignment

Connecting to Next Generation Science Standards (NGSS)

PE Code	Performance Expectation	DCI	CCC
HSS-ESS2-6	Develop a quantitative model to describe the cycling of carbon among the hydrosphere, atmosphere, geosphere, and biosphere.	ESS2.D: Weather and Climate - Gradual atmospheric changes were due to plants and other organisms that captured carbon dioxide and released oxygen. - Changes in the atmosphere due to human activity have increased carbon dioxide concentrations and thus affect climate.	Energy and Matter The total amount of energy and matter in closed systems is conserved.
HS-ESS3-6	Use a computational representation to illustrate the relationships among Earth systems and how those relationships are being modified due to human activity.	ESS2.D: Weather & Climate Current models predict that, although future regional climate changes will be complex and varied, average global temperatures will continue to rise. The outcomes predicted by global climate models strongly depend on the amounts of human-generated greenhouse gases added to the atmosphere each year and by the ways in which these gases are absorbed by the ocean and biosphere.	Systems and System Models When investigating or describing a system, the boundaries and initial conditions of the system need to be defined and their inputs and outputs analyzed and described using models.

Connecting to Computer Science Standards

California Computer Science Education Content Standards

[9-12S.AP.10](#). Algorithms & Programming.

Standard: Describe how artificial intelligence drives many software and physical systems.

AI4K12 AI's Five Big Ideas

Big idea #3: Learning

Concept: Nature of Learning. Describe how various types of machine learning algorithms learn by adjusting their internal representations.

Big idea #4: Natural interaction

Concept: Natural Language. Describe several approaches to natural language processing (e.g., how a chatbot can work using keywords, templates, or machine learning like GenAI).

Big idea #5: Societal Impact

Concept: Ethical AI. Articulate how to design an AI system in ethical ways.